A Deep Learning Model for Five Stage Diabetic Retinopathy Detection

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***Abstract*—Diabetic retinopathy (DR) is a common complication of diabetes that can result in vision loss if not diagnosed and treated in a timely manner. DR has several risk factors, including the duration of diabetes, poor control of blood sugar, hypertension, and high cholesterol levels. It is important for diabetics to undergo regular eye examinations in order to detect and manage this condition as early as possible. In order to slow down the progression of DR, healthcare professionals must be able to detect and classify the disease early in order to implement timely interventions and treatment strategies. Thus, it is possible to prevent the development of further visual impairments. This paper presents a novel approach for detecting and classifying DR stages based on image data. An array of retinal images obtained from diabetic patients was analyzed using state-of-the-art deep learning techniques. As a result, it aims to automate the detection and classification of Diabetic Retinopathy across its five stages. A pre-trained EfficientNetB3 model was applied with the transfer learning technique and achieved 90.03 percent accuracy. By employing this approach, this debilitating disease could be diagnosed and treated more effectively.**

***Index Terms*—Diabetic Retinopathy, Classification, Deep Learning, Transfer Learning.**

I. Introduction

Patients with diabetes may experience vision impairment or even blindness due to the DR. The symptoms of this condition impact the retina, the light sensitive tissue located at the back of the eye. There are several different types of retinal disorders that occur due to the damage of retinal blood vessels. It is often the case that DR does not present any symptoms in its early stages. Some individuals, however, may experience significant changes in their vision, such as difficulty reading or observing objects at a distance. As the condition progresses, the retinal blood vessels may become increasingly damaged, resulting in more serious complications. The figure illustrating a distinction between a healthy retina and the one that has been affected by Diabetic Retinopathy is shown in Figure 1 .

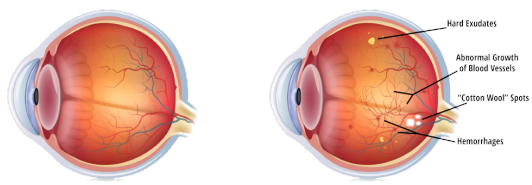

Fig. 1. Image of Normal Retina and DR affected Retina

As the disease progresses, these vessels may bleed into the vitreous, which fills the eye. As a result of this bleeding, individuals may notice dark spots floating in the air which resemble cobwebs. DR is primarily caused by diabetes, which is associated with consistently high blood sugar levels. The tiny blood vessels that supply the retina bleed or leak fluid when sugar levels are elevated. This blockage causes the retina to grow new blood vessels. This condition is further exacerbated by the fact that these new vessels are often weak and abnormal. This causes them to leak or bleed, thereby aggravating the condition. Maintaining a healthy blood sugar level and scheduling regular eye exams are essential to protecting the vision.

Prior to DR worsening, early detection is crucial to preventing further complications. The detection of diabetic retinopathy can be achieved using a variety of approaches, including manual and clinical examinations, including dilated eye exams, visual acuity testing, fundus photography, Optical Coherence Tomography (OCT), fluorescein angiography, and automated methods. Using new age automated techniques, which utilize the power of Deep Learning and Machine Learning, it is possible to achieve faster and more accurate results than ever before. These techniques are important because they manage large amounts of data, minimize human errors, and ensure reliable and consistent outcomes. By doing so, the detection process becomes more efficient and accessible. Moreover, these methods do not require any human resources, which makes them cost-effective. As a result, decisions can be made with greater accuracy and precision.

An objective of the proposed research is to improve the early diagnosis and treatment of Diabetic Retinopathy to help prevent visual impairment and blindness. Through automating the detection process, it is possible to detect Diabetic Retinopathy at an initial stage when the chances of treating it are high. To ensure timely treatment, we aim to develop an efficient and robust deep learning model for DR classification into multiple stages. To achieve accurate DR classification, a precise and efficient deep learning model has been developed. By training on large datasets of retinal images, the model learns to recognize patterns that signify different stages of Diabetic Retinopathy. The model has been tested and validated with promising results. Therefore, vision loss will be prevented as quickly and effectively as possible. The real-world application of this

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model includes its integration into clinical settings where it can assist ophthalmologists in identifying and classifying DR swiftly. This automated system can be deployed in remote and under-resourced areas, providing essential diagnostic support where specialized medical professionals may be scarce. Ultimately, the model aims to enhance patient outcomes by facilitating early intervention and tailored treatment plans.

This paper follows the following structure. Section II contains an overview of different studies on DR, the techniques used, and the results of those studies. The Section III describes the dataset used, the dataset's description, the proposed method, the pre-trained model employed, preprocessing and data augmentation steps, and at the end the training method. Section IV presents the results, including accuracy metrics, graphs, and a confusion matrix. Finally, Section V concludes the paper.

II. Literature Survey

There have been several studies that demonstrate the effectiveness of deep learning models in detecting early DR. Gulshan et al. [1] (2016) developed a CNN-based algorithm that attained high sensitivity and specificity in identifying DR from fundus images. Ratul Ghosh et al. [2] (2017) developed an improved CNN architecture that achieved 85% accuracy in multinomial classification and 95% accuracy in binary classification.

Using EfficientNet with XGBoost on the APTOS 2019 Blindness detection dataset, Yunika Laurensia et al. [3] achieved an accuracy rate of 98% for binary class classification. In spite of the fact that EfficientNet with XGBoost performed well on the APTOS 2019 Blindness detection dataset, it may not necessarily perform well on other datasets. The method should be evaluated using other datasets to assess its effectiveness.

M Dermksian et al. [4] (2021) used VGG16 on Gaussian Filtered dataset and achieved 80% accuracy for multinomial classification and 97% accuracy for binary classification. Abhay M Pamadi et al. [5] (2022) used transfer learning using MobileNetV2 with CNN on a Gaussian filtered dataset and achieved 78% accuracy for multinomial classification and 97% accuracy for binary classification.

Sambit S. Mondal et al. [6] (2022) used Ensemble deep learning techniques on APTOS 2019 blindness detection dataset which achieved 86.08% accuracy for multinomial classification. M. S. Ali Mazumder et al. [7] (2022) used MobileNetV2 on Gaussian filtered dataset achieved 96.04% for binary classification. Tawfiq Beghriche et al. [8] (2023) used Xception with multilayer fine-tuning on Gaussian filtered dataset achieved 97.95% for binary classification. Eshita Chowdhury et al. [9] (2023) used MobileNetV2 on Gaussian filtered dataset achieved accuracy of 97.82% for binary classification.

Based on the above reviewed studies, two key approaches are evident concerning DR classification. The first one is a yes and no sort of classification, which seeks to answer the common question, does this person have DR or not? But in this case, there is no answer to the stage of disease. The solution is multinomial classification, which classifies DR into five levels. This is important because a level four or five DR can be very serious. It is surprising that not much work has been done on multinomial classification which is the essence of dividing DR into five levels. This paper aims to contribute to this field by focusing on the aspects of developing a Deep Learning model for this classification. In addition, this model is applicable for creating mobile applications as well.

III. Methodology

A. Dataset

The dataset used is from the kaggle [10] named Diabetic Retinopathy 224x224 Gaussian Filtered Dataset. The complete dataset contains approximately 3662 images. The dataset comprises gaussian filtered retina scan photographs for the purposes of DR detection. Original dataset can be found at APTOS 2019 Blindness Detection [11] This image data is prepared in such a manner providing the dimensions of 224x224 so as to use it in number of existing pre-trained deep learning models comfortably.

Retinal imaging is a diagnostic procedure that captures detailed digital images of the inner back surface of the eye. This process, known as fundus photography, involves taking pictures of the eye's rear part, called the fundus. To do this, specialized cameras are used, which combine a sophisticated microscope with a camera that has a flash. Gaussian filtering is a technique used in image processing to reduce noise and smooth out images using Gaussian kernel. This method helps to blur out insignificant details and noise, making important features more prominent and improving the overall quality of the image. The five classes of the dataset and the number of images are provided in Table II

TABLE II
Distribution of classes and no.of images in the dataset.

	Classname	No.of Images
0	Mild	370
1	Moderate	999
2	No_DR	1805
3	Proliferate_DR	295
4	Severe	193

B. Proposed Method

1) Pre-trained Model: As for image detection, there are a number of pre-trained models available, such as MobileNet, VGG16, Inception-ResNet, Xception, EfficientNet, ResNet, and DenseNet. Out of these, EfficientNet was observed to be the most suitable architecture for this dataset. Out of several pre-trained models put to test under the research, EfficientNet remained the most successful. According to empirical studies, Networks based on EfficientNet architecture can successfully compete with VGG-16, Inception V3, ResNet-50, MobileNet,

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TABLE I
OVERVIEW OF THE EXISTING WORKS IN DETECTING DIABETIC RETINOPATHY

Paper Title	**Year**	**Techniques Used**	**Dataset**	**Accuracy**	**Classification Type**
Automatic detection and classification of diabetic retinopathy stages using CNN	2017	New CNN architecture	Diabetic Retinopathy Detection	85%	MC
				95%	BC
Early Detection of Diabetic Retinopathy Cases using Pre-trained EfficientNet and XGBoost	2020	EfficientNet + XGBoost	APTOS 2019 Blindness Detection	98%	BC
Diabetic Retinopathy Detection via CNN	2021	VGG16	Diabetic Retinopathy 224x224 Gaussian Filtered	80%	MC
				97%	BC
Diabetic Retinopathy Detection using MobileNetV2 Architecture	2022	Transfer Learning MobileNetV2 and CNN	Diabetic Retinopathy 224x224 Gaussian Filtered	78%	MC
				97%	BC
An Ensemble Deep Learning Technique for Detection and Classification of Diabetic Retinopathy	2022	CLAHE- Preprocessing, GAN- Data augmentation, Ensemble of modified DenseNet101 and ResNeXt	APTOS19	86.08%	MC
				96.98%	BC
Deep Learning Approaches for Diabetic Retinopathy Detection by Image Classification	2022	MobileNetV2	Diabetic Retinopathy 224x224 Gaussian Filtered	96.04%	BC
A multi-level fine-tuned deep learning based approach for binary classification of diabetic retinopathy	2023	Xception with multilayer finetuning	Diabetic Retinopathy 224x224 Gaussian Filtered	97.95%	BC
Deep Learning-based Diabetic Retinopathy Detection for automated health assessment	2023	MobileNetV2	Diabetic Retinopathy 224x224 Gaussian Filtered	97.82%	BC

*MC represents Multinomial Classification and *BC represents Binary Classification

and several other popular architectures on many image recognition tasks utilizing transfer learning.

EfficientNet is a set of CNN designs that focus on optimizing the accuracy and efficiency of a model by controlling its depth, width, and resolution. It consists of eight variants: B0, B1, B2, B3, B4, B5, B6, and B7, where size and complexity increase with the number. The least complex of them is B0, whereas among them, B7 has the most layers and obviously more complexity overall. Like MobileNetV2 and MnasNet, the architecture utilizes MBConv, but on a larger scale. While experimenting with different variants of EfficientNet included in the study, B3 variant suited the dataset very well. This model customization capability, in particular, will ensure high accuracy and its effective execution in image tasks.

2) Preprocessing and Data Augmentation: As the dataset is imbalanced, the Synthetic Minority Over-sampling Technique (SMOTE) was used for data augmentation. Applying SMOTE significantly improved the model's performance by reducing bias towards the majority class and increasing accuracy in detecting minority class instances. After applying SMOTE, the dataset was balanced with 1805 images in each class. To address the wide variation in pixel intensity values in retinal images, which can lead to biased learning, the Standard Scaler was applied. This step ensured that pixel intensity values were standardized, allowing each feature to contribute equally during the model's training process.

Feature extraction utilizing Principal Component Analysis (PCA) was required to handle the high-dimensional pixel level retinal images. So, instead of having high-dimensioned data, PCA algorithms create better and less complex features. A graph that plotted the percentage of variance explained against the number of components sequentially added helped to find a good proportion of the total complexity retained in the dataset while still being able to limit the number of components. Between three thousand to three thousand five hundred components were found to be effective for this dataset and further analysis led to three thousand four hundred sixty-eight being the most effective. This not only reduces the

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training time but also enhances the performance of the model and reduces the chances of overfitting the model.

3) Training method: The neural networks for the diabetic retinopathy dataset, consisting of 224x224 Gaussian filtered images, are trained using transfer learning technique. The output of PCA is fed as input to the architecture. The pre-trained model leverages learned features from a large and diverse dataset, capturing essential patterns and structures relevant to image analysis. The weights of EfficientNetB3, pre-trained on the ImageNet dataset, are loaded serving as the core feature extractor in this architecture. The proposed model architecture combines PCA for dimensionality reduction with the advanced feature extraction capabilities of EfficientNetB3. It employs a carefully structured sequence of dense layers with ReLU activation, includes a dropout layer for regularization, and features a final softmax output layer for multi-class classification.

A decent optimizer has the ability to enhance the entire training activity,ensure that the model does not get stuck on bad local optima and eventually, bring great training results. In this example, the Adam optimizer is used to achieve the best result in minimizing categorical cross-entropy loss. Due to its adaptive learning rate and its effectiveness, it is frequently used in deep learning applications. For preventing overfitting which is likely to happen when many epochs when models are built, a callback feature is incorporated. These systems are introduced to minimize time taken to train the models and time taken to test the models, in order for the model to perform optimally on the diabetic retinopathy dataset.

IV. Experimental Evaluation

Initially, the dataset was tested with several architectures, including DenseNet121, InceptionResNetV2, EfficientNetB3, VGG16, MobileNetV2, InceptionV3, and ResNet50. This preliminary testing was conducted to identify the most suitable architecture. Comparing different architectures, EfficientNetB3 worked better for this dataset. These are accuracies of the models trained using the initial dataset (without applying PCA). Model architectures and their accuracies are given in Table III

TABLE III
Comparision of Model Accuracies

Model	Train_Accuracy	Test_Accuracy	Val_Accuracy
EfficientNetB3	83.01%	79.48%	75.82%
Densenet121	86.89%	73.25%	79.64%
InceptionResnetV2	90.26%	72.60%	75.27%
VGG16	95.95%	73.36%	74.91%
MobileNetV2	72.68%	73.25%	71.09%
InceptionV3	65.35%	60.48%	60.55%
ResNet50	71.08%	71.07%	72.91%

After selecting EfficientNet, its seven variants were trained on the final dataset, which had PCA applied. Out of them B3 variant was working well for this dataset. As mentioned earlier, the Adam optimizer and categorical cross-entropy loss, with a batch size of 32, were used for training. And the results are mentioned in Table IV. The performance measures of the proposed model are also mentioned in Table V.

TABLE IV
Model Accuracies

Training Accuracy	94.74%
Testing Accuracy	90.03%
Validation Accuracy	89.75%

TABLE V
Performance measures of different stages of Diabetic Retinopathy

	Precision	Recall	F1-Score
Mild	0.79	0.94	0.86
Moderate	0.87	0.89	0.88
No_DR	0.95	0.86	0.90
Proliferate_DR	0.96	0.86	0.91
Severe	0.96	0.96	0.96

Other metrics such as Model accuracy, Model loss, Model AUC Graphs, Confusion matrix, ROC-AUC Curves are also considered.

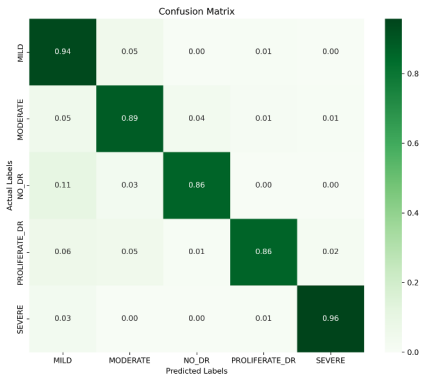

Fig. 2. Confusion Matrix

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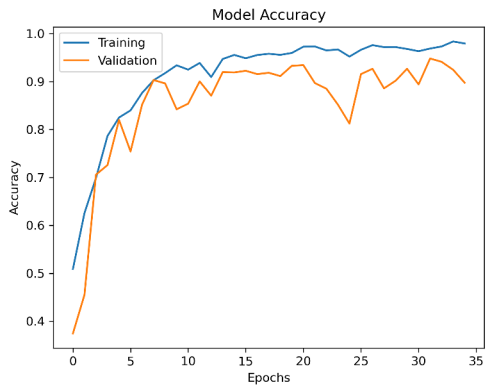

Fig. 3. Model Accuracy Graph

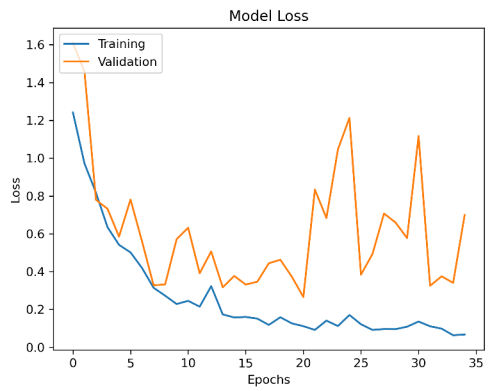

Fig. 4. Model Loss Graph

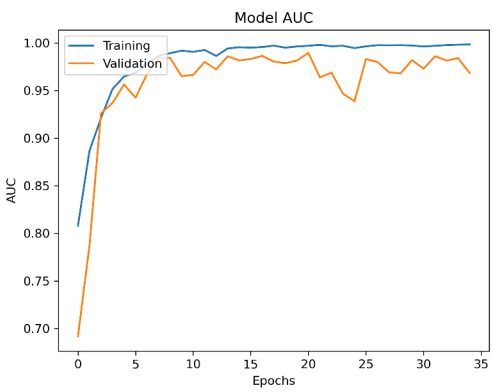

Fig. 5. Model AUC Graph

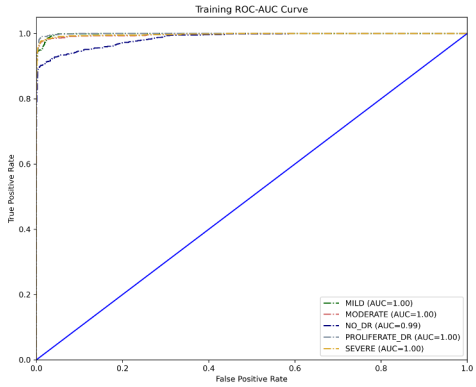

Fig. 6. Training ROC-AUC Curve

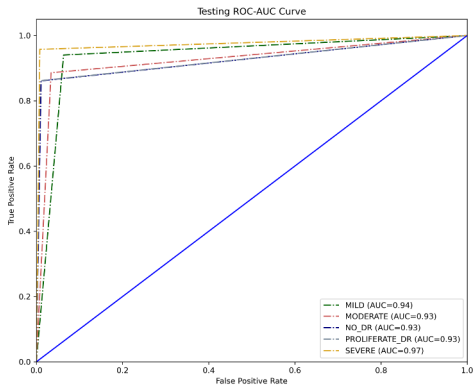

Fig. 7. Testing ROC-AUC Curve

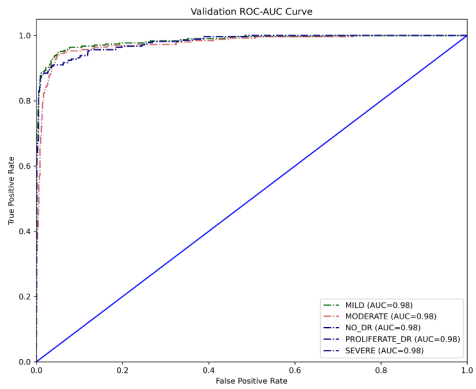

Fig. 8. Validation ROC-AUC Curve

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V. Conclusion

This study shows how effective transfer learning and deep learning techniques are used in classifying diabetic retinopathy into different stages. DR is a serious eye condition caused by diabetes which can cause vision impairment if it's not detected and treated promptly. The proposed model with EfficientNetB3 was the best, achieving high testing accuracy of 90.03%. Using SMOTE for data augmentation and PCA for feature extraction greatly improved the model's performance. This research highlights the potential of deep learning models for early diagnosis and classification of Diabetic Retinopathy into five stages, with practical applications in small-scale systems like mobile apps. These tools could play a crucial role in preventing severe vision impairment through timely diagnosis and management.

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